ELL715 - Assignment 4

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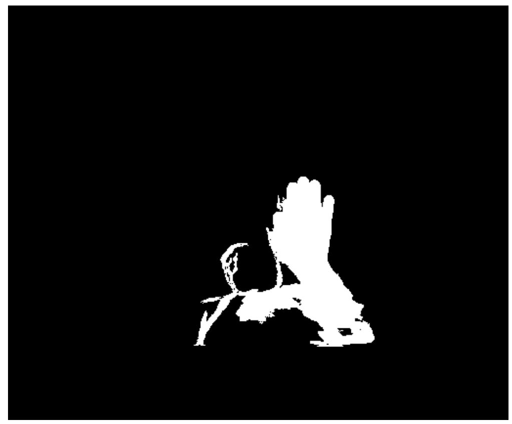
1 Part1: Paper Implementation

1.1 Implementation of Silhouette Tunnel

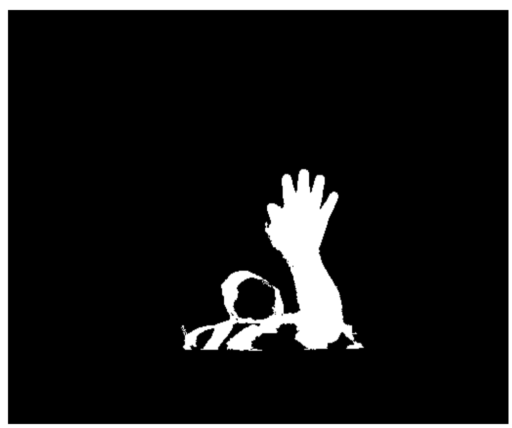
(a) original-Compass



(b) silhouette mask - Compass

Figure 1: Extraction of silhouette mask from compass

(a) original-Piano



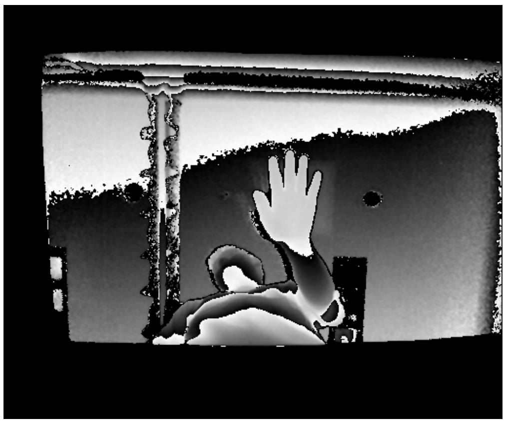
(b) silhouette mask - Piano

Figure 2: Extraction of silhouette mask from piano

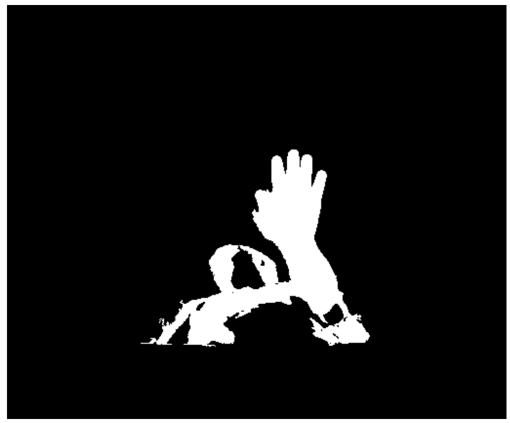
(a) original-Push

(b) silhouette mask - Push

Figure 3: Extraction of silhouette mask from Push



(a) original-UCDO



(b) silhouette mask - UCDO

Figure 4: Extraction of silhouette mask from UCDO

Each pixel within the silhouette tunnel is represented by a 14-dimensional feature vector:

$$\mathbf{f} = \left[x,\, y,\, t,\, z,\, d_E,\, d_W,\, d_N,\, d_S,\, d_{NE},\, d_{NW},\, d_{SE},\, d_{SW},\, dT^+,\, dT^-\right]^T$$

where:

- x, y, t denote the spatial and temporal coordinates of the pixel,
- z is the depth value,
- $d_E, d_W, d_N, d_S, d_{NE}, d_{NW}, d_{SE}, d_{SW}$ represent directional distances from the pixel to the nearest background boundary in the eight cardinal directions, and
- dT^+, dT^- denote the forward and backward temporal persistence of the silhouette pixel across time.

Directional distances and temporal runs are efficiently computed using Numba-accelerated kernels for parallel processing.

All feature vectors from the sequence are collected into a feature matrix:

$$\mathbf{F} = \begin{bmatrix} \mathbf{f}_1 & \mathbf{f}_2 & \dots & \mathbf{f}_N \end{bmatrix} \in \mathbb{R}^{14 \times N}$$

and are normalized feature-wise to the range $[0, 1]$ as:

$$\mathbf{F}_{\text{norm}} = \frac{\mathbf{F} - \mathbf{F}_{\text{min}}}{\mathbf{F}_{\text{max}} - \mathbf{F}_{\text{min}} + \epsilon}$$

where ϵ is a small constant (e.g., 10^{-6}) to avoid division by zero.

1.2 Extraction of Covariance Matrix

The silhouette tunnel provides us with a rich representation of the pixel features over time. To analyze the variability and relationships between these features, we compute the covariance matrix of the normalized feature matrix:

$$\mathbf{C} = \frac{1}{N-1} \mathbf{F}_{\text{norm}} \mathbf{F}_{\text{norm}}^T$$

where $\mathbf{C} \in \mathbb{R}^{14 \times 14}$ is the covariance matrix capturing the pairwise feature correlations across all pixels in the silhouette tunnel across all the frames.

Note: In the paper, the denominator of the above equation is given as N. However, to obtain an unbiased estimate of the covariance, we use $N - 1$ in our implementation.

We then just take the upper triangular part of the covariance matrix (including the diagonal) and vectorize it to form our final feature descriptor for the video sequence:

$$\mathbf{d} = \text{vec}(\mathbf{C}_{\text{upper}})$$

where $\mathbf{d} \in \mathbb{R}^{105}$ is the final feature descriptor capturing the relationships between the pixel features over time.

1.3 Temporal Hierarchical Covariance Descriptor

To capture the temporal dynamics of each gesture, the feature matrix $F \in \mathbb{R}^{14 \times N}$ is divided into multiple temporal segments. We use a three-level hierarchy where the sequence is split into $1 + 2 + 4 = 7$ parts in total, giving both coarse and fine temporal resolution.

For each segment, a covariance matrix is computed as:

$$C = \frac{1}{N-1} \sum_{n=1}^{N} (f_n - \mu)(f_n - \mu)^\top, \quad \text{where} \quad \mu = \frac{1}{N} \sum_{n=1}^{N} f_n$$

Each C is symmetric, so we only keep its upper-triangular entries (105 values for 14 features). The seven such vectors are then concatenated to form the final descriptor:

$$\text{Descriptor}_{TH} = [C_1^{ut}, C_2^{ut}, \dots, C_7^{ut}] \in \mathbb{R}^{735}$$

This representation captures how feature correlations evolve over time while remaining compact. **Implementation Note:** all covariance computations are parallelized using Numba for efficiency.

Result Note: There has been barely an improvement in the results after implementing temporal hierarchy. The table showing the results before and after implementing temporal hierarchy is as follows:

Gesture	Without Temporal Hierarchy	With Temporal Hierarchy
Compass	0.280	0.289
Piano	0.305	0.303
Push	0.280	0.275
UCDO	0.298	0.275

Table 1: Comparison of Recognition Accuracy With and Without Temporal Hierarchy using EER

The histograms can be seen in the results/TemporalHierarchy/ folder, and also the Figure 9b and the following figures.

1.4 Hand-Morphology:

For each sub-silhouette, a separate feature matrix $F_k \in \mathbb{R}^{14\times N_k}$ is computed, following the same 14-dimensional feature representation used for the full silhouette. Each sub-silhouette then produces its own temporal-hierarchical covariance descriptor $C_k \in \mathbb{R}^{735}$.

The final multichannel descriptor is formed by concatenating the descriptor of the full silhouette with those of the $K = 3$ sub-silhouettes:

$$\text{Descriptor}_{multi} = [\, C_{\text{full}},\, C_1,\, C_2,\, C_3 \,]$$

This results in a combined representation of dimension:

$$4 \times 735 = 2940$$

Some examples of subsilhouettes representation:

(a) Full Silhouette = **(b)** Sub-silhouette 1 + **(c)** Sub-silhouette 2 + **(d)** Sub-silhouette 3

Figure 5: Example of sub-silhouette decomposition from a full silhouette for the compass gesture.

(a) Full Silhouette = **(b)** Sub-silhouette 1 + **(c)** Sub-silhouette 2 + **(d)** Sub-silhouette 3

Figure 6: Example of sub-silhouette decomposition from a full silhouette for the piano gesture.

(a) Full Silhouette = **(b)** Sub-silhouette 1 + **(c)** Sub-silhouette 2 + **(d)** Sub-silhouette 3

Figure 7: Example of sub-silhouette decomposition from a full silhouette for the push gesture.

(a) Full Silhouette = **(b)** Sub-silhouette 1 + **(c)** Sub-silhouette 2 + **(d)** Sub-silhouette 3

Figure 8: Example of sub-silhouette decomposition from a full silhouette for the UCDO gesture.

Note: The above study has been done for test sample 001 of user 02, 2nd frame, considering the 1st frame as background; for all the gestures, i.e., Compass, Piano, Push and UCDO.

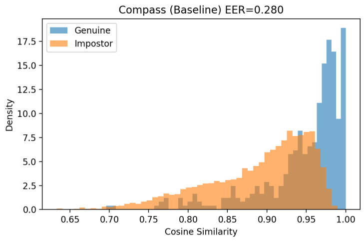
For cosine similarity:

$$d_{\text{cosine}} = 1 - \frac{1}{4} \sum_{k \in \{\text{full},1,2,3\}} \frac{\langle C_k^{(a)}, C_k^{(b)} \rangle}{\|C_k^{(a)}\|_2 \, \|C_k^{(b)}\|_2}$$

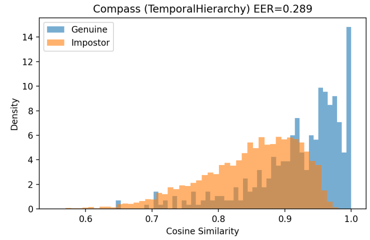
where $C_k^{(a)}$ and $C_k^{(b)}$ are the covariance descriptors of the k-th channel for samples a and b.

1.5 ablation study

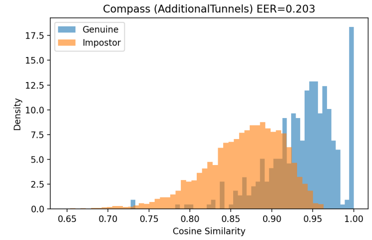
We have considered three scenerios for the ablation study, which includes; Baseline, Baseline + Temporal Hierarchy, and Baseline + Temporal Hierarchy + Hand Morphology. The results are as follows:



(a) Baseline

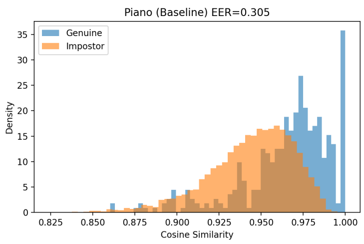


(b) Baseline + Temporal Hierarchy

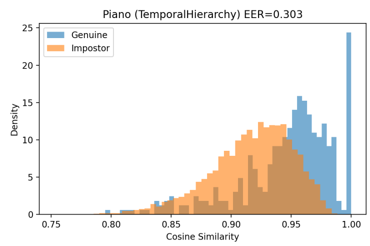


(c) Baseline + Temporal Hierarchy + Hand Morphology

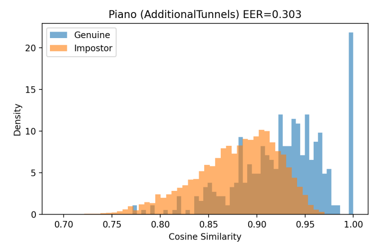
Figure 9: Ablation Study Confusion Matrices for Different Scenarios



(a) Baseline

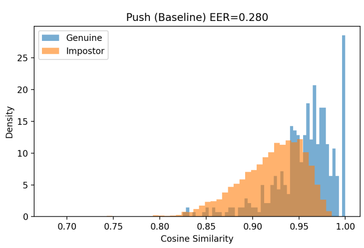


(b) Baseline + Temporal Hierarchy

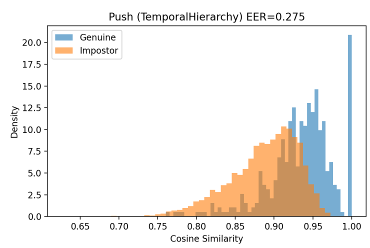


(c) Baseline + Temporal Hierarchy + Hand Morphology

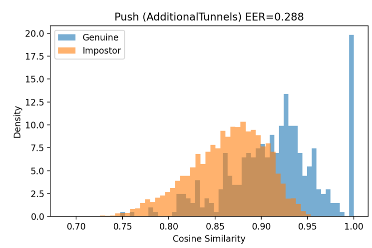
Figure 10: Ablation Study Confusion Matrices for Different Scenarios - Piano Gesture



(a) Baseline

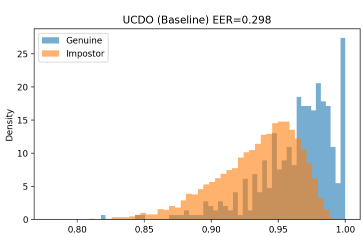


(b) Baseline + Temporal Hierarchy

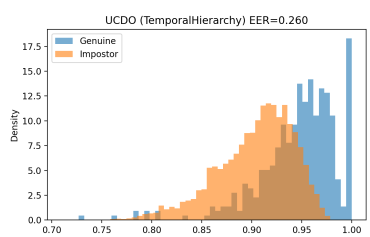


(c) Baseline + Temporal Hierarchy + Hand Morphology

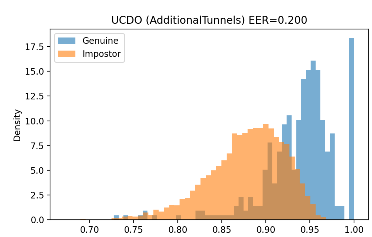
Figure 11: Ablation Study Confusion Matrices for Different Scenarios - Push Gesture



(a) Baseline



(b) Baseline + Temporal Hierarchy



(c) Baseline + Temporal Hierarchy + Hand Morphology

Figure 12: Ablation Study Confusion Matrices for Different Scenarios - UCDO Gesture

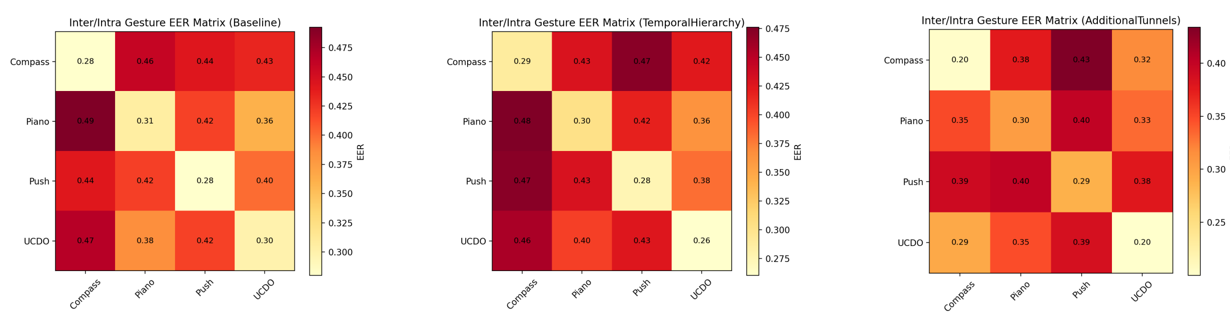
Table 2: Equal Error Rate (EER) comparison across different ablation settings and gestures.

Gesture	Baseline	Temporal Hierarchy	Hand Morphology
Compass	0.280	0.265	0.204
Piano	0.305	0.303	0.280
Push	0.260	0.275	0.200
UCDO	0.299	0.260	0.226

1.6 Impact of different gestures, and intra and inter-gesture matching:

1.6.1 Comparison of Inter-Gesture and Intra-Gesture Mappings:

Intra-Gesture Matching: This involves comparing samples of the same gesture type to evaluate consistency within that gesture. For example, comparing different instances of the "Compass" gesture to see how similar they are to each other.
Inter-Gesture Matching: This involves comparing samples of different gesture types to assess how distinct each gesture is from the others. For example, comparing "Compass" gestures against "Piano" gestures to evaluate their separability.
EER Matrices: The matrix consists of EER values where each entry (i, j) represents the EER when comparing gesture type i against gesture type j. The diagonal entries represent intra-gesture EERs (same gesture comparisons), while the off-diagonal entries represent inter-gesture EERs (different gesture comparisons).



(a) Baseline **(b)** Temporal Hierarchy **(c)** Hand Morphology

Figure 13: Gesture EER Matrices for Different Variants

Table 3: Gesture EER Matrix - Baseline

	Compass	Piano	Push	UCDO
Compass	0.28	0.46	0.44	0.43
Piano	0.49	0.31	0.42	0.36
Push	0.44	0.42	0.28	0.40
UCDO	0.47	0.38	0.42	0.30

Table 4: Gesture EER Matrix - Temporal Hierarchy

	Compass	Piano	Push	UCDO
Compass	0.29	0.43	0.47	0.42
Piano	0.48	0.30	0.42	0.36
Push	0.47	0.43	0.28	0.38
UCDO	0.46	0.40	0.43	0.26

Table 5: Gesture EER Matrix - Hand Morphology

	Compass	Piano	Push	UCDO
Compass	0.20	0.38	0.43	0.32
Piano	0.35	0.30	0.40	0.33
Push	0.39	0.40	0.29	0.38
UCDO	0.29	0.35	0.39	0.20

To analyze the influence of gesture type on authentication performance, both **intra-gesture** and **inter-gesture** evaluations were conducted. Let G_i denote a *gallery gesture* and G_j denote a *probe gesture*, where $i, j \in \{1, 2, 3, 4\}$ correspond to the gestures {Compass, Piano, Push, UCDO}.

The evaluation protocol proceeds as follows:

Step 1: Gallery Construction: For each gesture G_i, a gallery template is created for every enrolled subject ($s_g \in \{01, \dots, 08\}$) by averaging the descriptors of all available trials:

$$\mathbf{D}^{(i)}_{\text{gallery},s_g} = \frac{1}{N_i} \sum_{k=1}^{N_i} \mathbf{d}^{(i,k)}_{s_g}$$

where $\mathbf{d}^{(i,k)}_{s_g}$ represents the k-th descriptor for gesture G_i of subject s_g. Note that this is different from the protocol followed above, where only one sample was used at a time for gallery, and all the combinations were considered while computing genuine and impostor scores.

Step 2: Probe Samples: Probe samples are taken from subjects $s_p \in \{09, \dots, 16\}$ performing gesture G_j. Each probe descriptor is denoted $\mathbf{d}^{(j,m)}_{s_p}$.

Step 3: Score Computation: For every gallery–probe pair, the cosine similarity is computed as:

$$\text{sim}\left(\mathbf{D}^{(i)}_{\text{gallery},s_g}, \mathbf{d}^{(j,m)}_{s_p}\right) = 1 - \text{cosine\_distance}\left(\mathbf{D}^{(i)}_{\text{gallery},s_g}, \mathbf{d}^{(j,m)}_{s_p}\right)$$

Genuine and impostor scores are defined as:

$$\begin{cases} s_g = s_p, & \text{Genuine scores (same subject)} \\ s_g \neq s_p, & \text{Impostor scores (different subjects)} \end{cases}$$

Step 4: EER Computation: For each gesture pair (G_i, G_j), the Equal Error Rate (EER) is computed from the genuine and impostor score distributions.

- $i = j$: *Intra-gesture EER* — same gesture for gallery and probe.
- $i \neq j$: *Inter-gesture EER* — different gestures for gallery and probe.

Step 5: Gesture EER Matrix: The process yields a 4×4 EER matrix:

$$\text{EER}_{i,j} = f(G_i, G_j)$$

where diagonal elements ($i = j$) represent intra-gesture performance, and off-diagonal elements ($i \neq j$) measure inter-gesture confusion.

Several patterns, trennds and observations that appear from the aove results are:

1. Same-gesture matching is better.
Diagonal entries of the EER matrices (intra-gesture) are substantially lower than the off-diagonal ones. This indicates that when the same gesture is used during both enrollment and testing, authentication accuracy is high. In contrast, cross-gesture (inter-gesture) matching produces significantly higher EERs, confirming that the descriptors are strongly gesture-sensitive.

2. Hand morphology improves performance.
The inclusion of sub-silhouette tunnels (morphological features) provides the largest reduction in diagonal EERs for certain gestures—particularly *Compass* and *UCDO*. This suggests that morphology-based cues (hand shape and structural consistency) add stability and complement the temporal-motion information in the descriptor.

3. Temporal hierarchy gives modest gains.
Dividing a gesture into temporal segments captures the dynamic structure of motion and yields small improvements in some gestures. However, the overall effect is less pronounced than the morphology-based approach, indicating that temporal segmentation alone cannot fully model gesture variability.

4. Inter-gesture confusion remains high.
Off-diagonal EER values remain elevated across all variants. This indicates that descriptors trained for one gesture do not generalize well to other gestures. In other words, the system is not gesture-agnostic and relies on users performing the same gesture during both enrollment and authentication.

It is also observed that Compass and UCDO gestures perform better than Piano and Push gestures. This could be due to the distinct hand shapes and motion patterns involved in Compass and UCDO, which are better captured by the morphology features. In contrast, Piano and Push gestures may involve more subtle or similar hand configurations, leading to higher confusion rates.

Will this system work for random hand gestures?
Not really. The observed results indicate that the descriptors we are using are gesture-dependent and requires consistent gestures for authentication. This will be further proved in an experiment which is mentioned next. But the current results indicate:

- **Gesture dependency:** The feature representation includes spatial coordinates (x, y, t), depth, and temporal run-lengths. These are strongly tied to the motion and dynamics of specific gestures. Random gestures will produce different patterns that do not match enrolled templates.

- **High inter-gesture EERs:** The elevated off-diagonal EERs (0.35–0.50 range) confirm that cross-gesture matching yields poor authentication accuracy.

1.7 Experiment 2: Multi-Gesture Enrollment Authentication

Objective

We try to analyze if a multi-gesture enrollment strategy can improve performance compared to single-gesture enrollment.
Instead of enrolling each user with a single gesture, we construct a universal gallery template by averaging feature descriptors from multiple gestures performed by the gallery subjects. Probe samples are then drawn from all gestures of probe subjects.

Methodology

For each variant (*Baseline*, *Temporal Hierarchy*, and *Hand Morphology*), the following protocol was used:

1. Average all descriptors from all gallery subjects (1–8) across all gestures.
2. A **universal gallery template** formed by averaging all gallery descriptors:

$$\bar{C}_{gallery} = \frac{1}{N_G} \sum_{i=1}^{N_G} C_i$$

   where C_i represents the descriptor vector from the i-th gallery sample.
3. Each probe descriptor from probe subjects (9–16) was compared against the universal template using cosine similarity.
4. Genuine scores were obtained from gallery descriptors (same population as enrolled users), while impostor scores were obtained from probe descriptors (unseen subjects).
5. The Equal Error Rate (EER) was computed from the two distributions.

Results

A summary of the multi-gesture enrollment performance for all three variants is shown below.

Table 6: Multi-Gesture Enrollment EER for Each Variant

Variant	EER (%)	Remarks
Baseline	0.27	Gesture-specific covariance only
Temporal Hierarchy	0.23	Captures temporal structure across segments
Hand Morphology (Additional Tunnels)	0.20	Adds sub-silhouette morphology cues

Interpretation

The results show that multi-gesture enrollment improves the overall system reliability. By averaging across gestures, the universal template becomes less sensitive to gesture-specific dynamics and captures more stable, user-specific hand traits.

- The **Hand Morphology** variant achieved the lowest EER (0.20), indicating that sub-silhouette cues provide complementary, gesture-invariant information.
- The **Temporal Hierarchy** variant improved over the Baseline by modeling dynamic consistency across temporal segments.
- The **Baseline** variant, which relies solely on covariance of full-silhouette features, remained the most gesture-dependent and hence less generalizable.

Conclusion

This experiment demonstrates that aggregating feature descriptors from multiple gestures can produce a more robust enrollment representation. However, while the average template reduces gesture sensitivity, inter-gesture variability remains significant, suggesting that morphology-based features (such as sub-silhouette representations) are more effective for gesture-invariant user authentication.

References